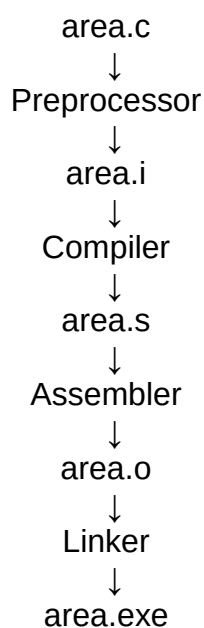
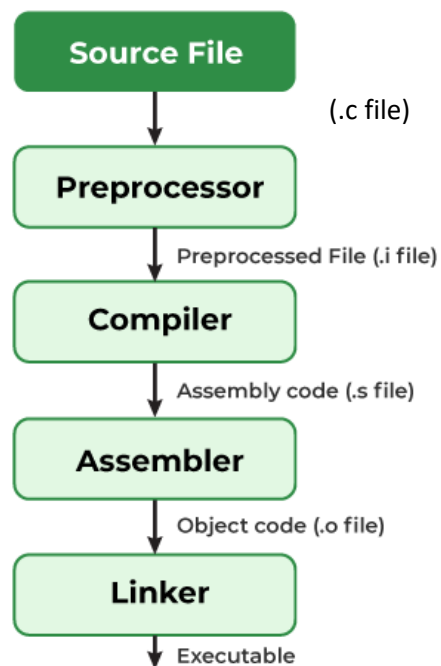


Compilation and Execution Process



Pre-Processing

All the statements starting with the **#** symbol in a C program are processed by the pre-processor.

i. Comments and whitespace are removed

```

/* This is a
multi-line comment in C */
#include<stdio.h>
int main()
{
    // this is a single-line comment in C
    return 0;
}

```

ii. Macros Expansion

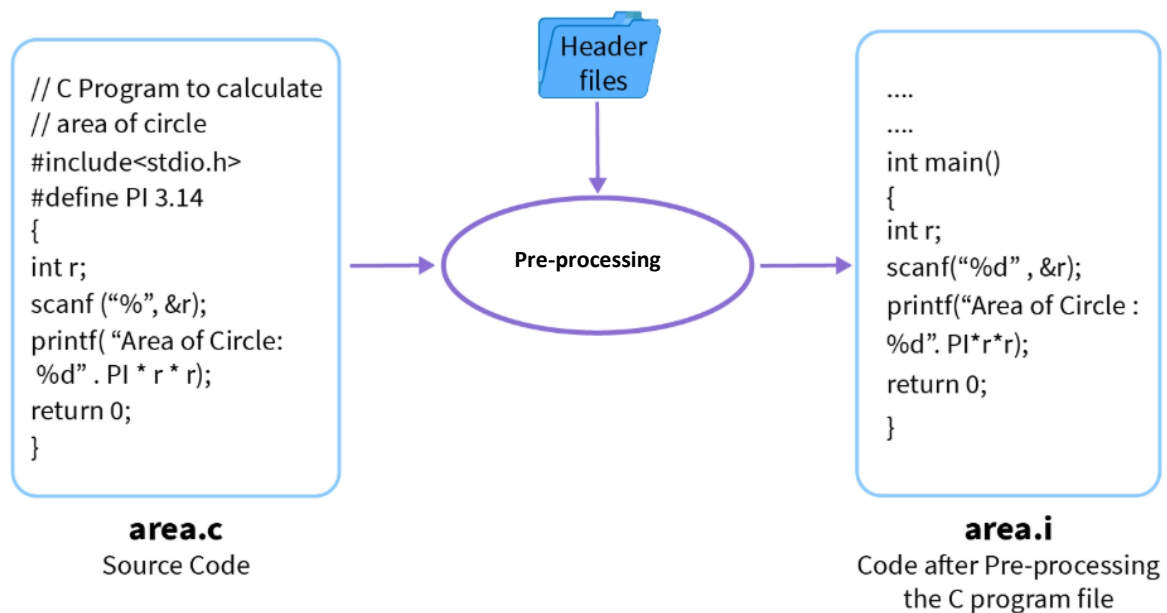
```
#define G 9.8
```

iii. File inclusion

```
#include <stdio.h>
```

```
#include<math.h>
```

```
#include<filename>
```



The Compilation Process in C

Compiling phase in C uses inbuilt compiler software to convert the intermediate (.i) file into an Assembly file (.s)

```

....
....
int main()
{
int r;
scanf("%d", &r);
printf("Area of Circle :
%d". PI*r*r);
return 0;
}

```

area.i

Code after Pre-processing
the C program file

Compiler Software

```

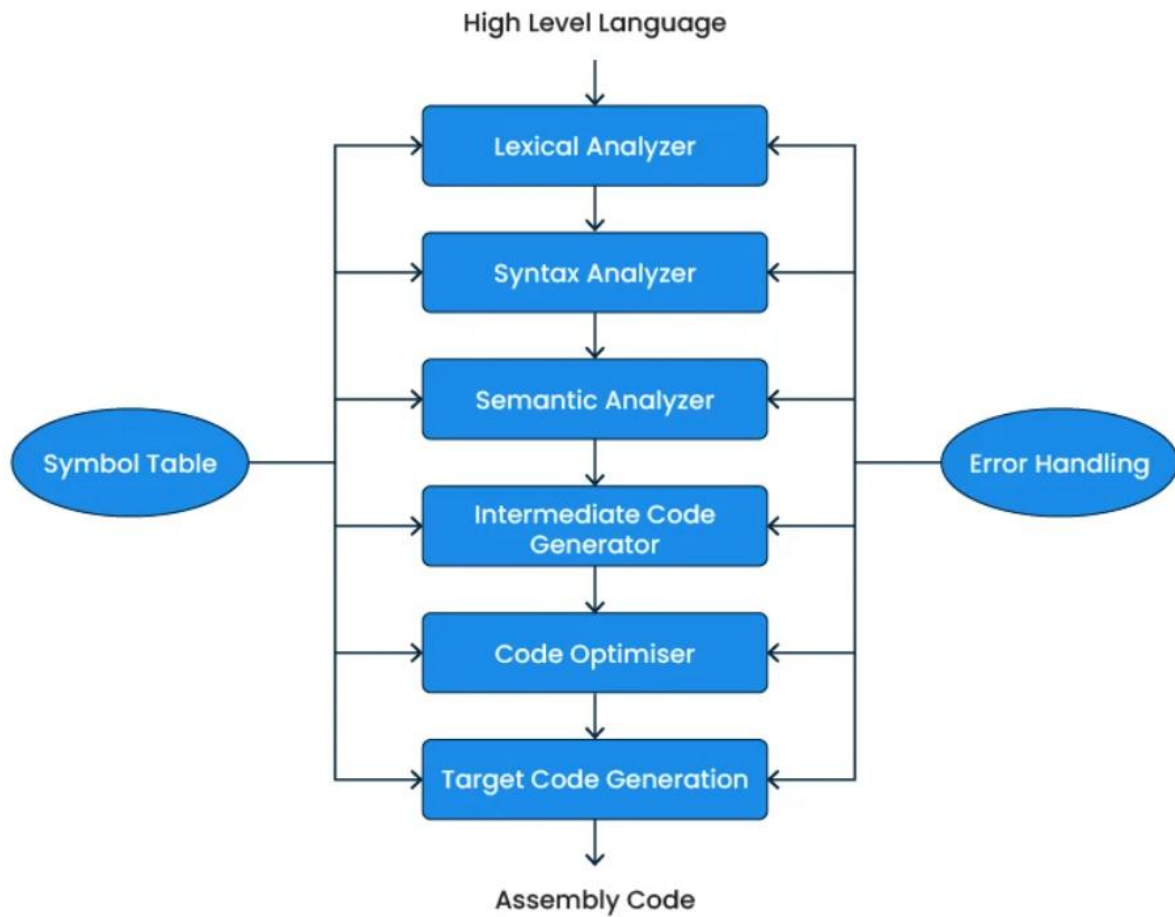
push ebp
mov ebp, esp
and esp, -16
...
ret

```

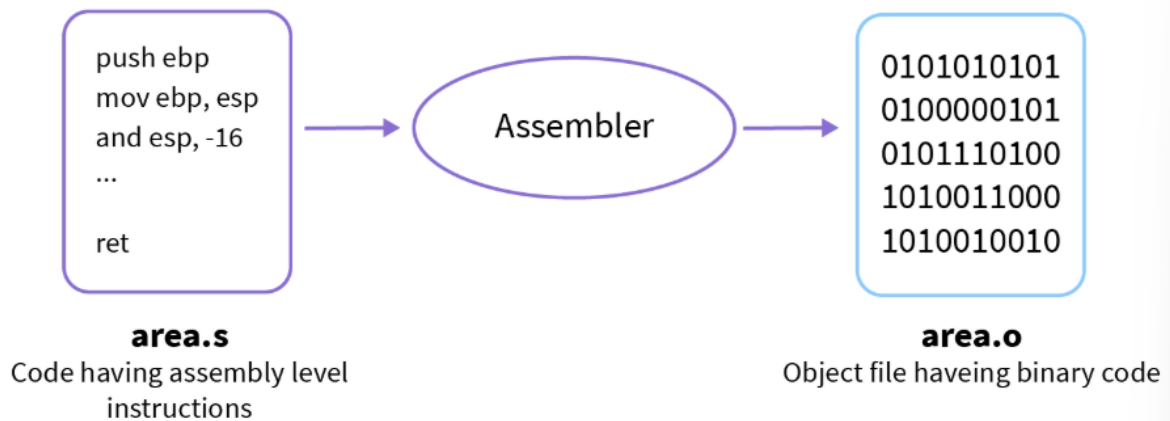
area.s

Code having assembly level
instructions

Phases of Compiler

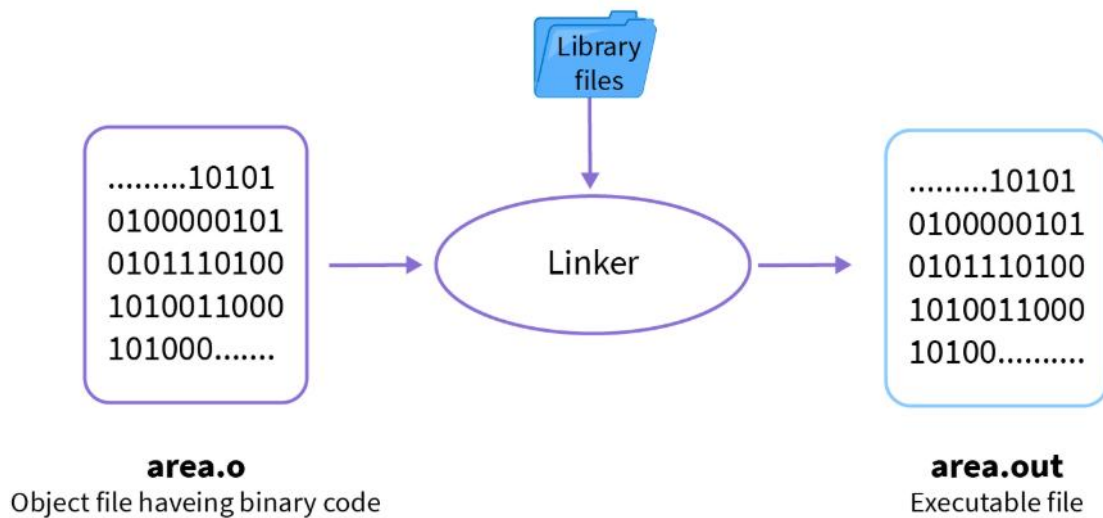


Assembler



Linker

Linking is a process of including the library files into our program. Library Files are some predefined files that contain the definition of the functions in the machine language and these files have an extension of .lib



Loader

Loader is a system program that loads the executable file into memory and starts its execution.